

# Statistics for Genomic Data Science: Week 3

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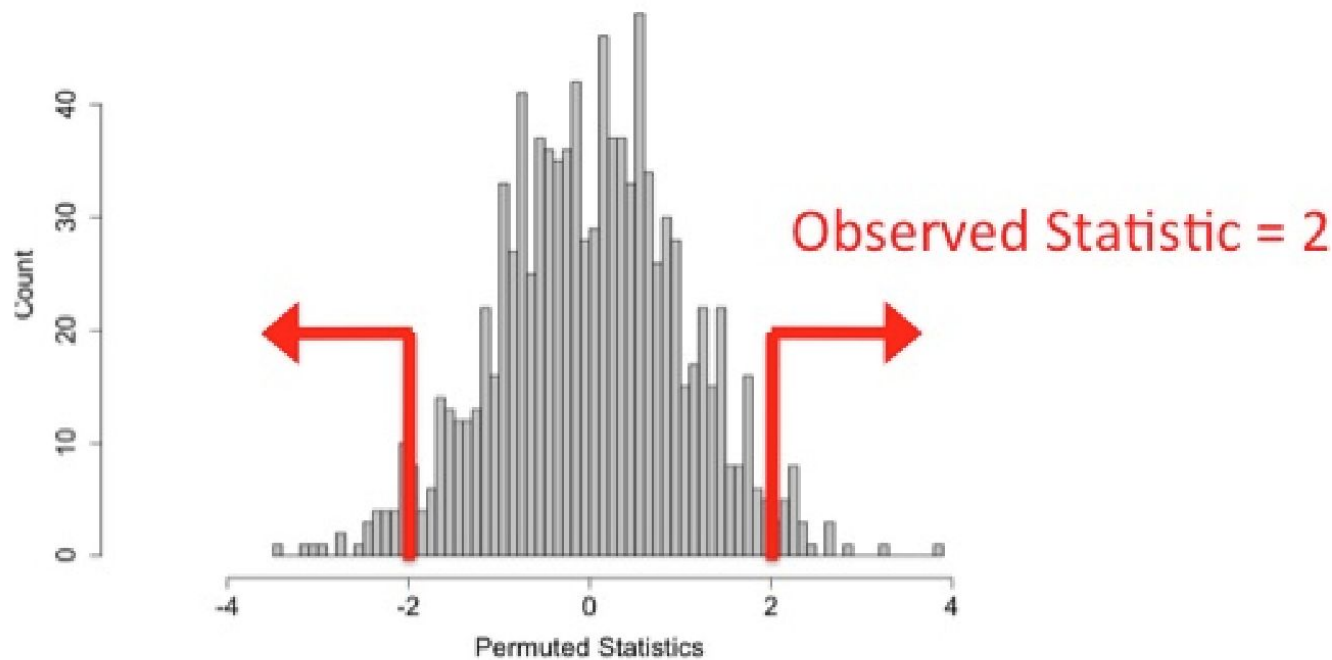
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**This week:**

**More modeling**

**Statistical significance**

**Multiple testing**



$$\{ \# \mid S^{perm} \mid \geq \mid S^{obs} \mid \}$$

$$\text{P-value} = \frac{\quad}{\# \text{ of Permutations}}$$

- Family wise error rate:

$$\Pr(\# \text{ False Positives} \geq 1)$$

- False discovery rate:

$$\mathbb{E} \left[ \frac{\# \text{ False Positives}}{\# \text{ Of Discoveries}} \right]$$

- EFP (e-values)

$$\mathbb{E}[\# \text{ False Positives}]$$